ModelIt!

Teacher's Guide

Why Does It Feel Colder When It's Windy?

Modeling Wind Chill and Convective Heat Loss

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-PS1-4, MS-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-4, MS-PS3-4

Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how wind speed and air temperature interact to drive convective heat loss, skin temperature changes, and perceived cold.
Disciplinary Core Idea	**PS3.A Definitions of Energy / PS3.B Conservation of Energy and Energy Transfer** — Energy is transferred from warmer to cooler objects or regions by conduction, convection, and radiation. Wind increases convective transfer rate, removing thermal energy from the skin faster.
Crosscutting Concept	**Cause and Effect** — Students investigate the causal mechanisms by which wind speed increases heat loss rate, identifying that the disruption of the thermal boundary layer is the key mechanism linking wind to perceived cold.

CAST Testing Alignment

MULTIPLE CHOICE:

Two students stand outside on a 35°F day. Student A is in a sheltered area with no wind. Student B is on a hilltop with 25 mph wind. Based on the wind chill model, which student will experience a lower skin temperature, and what is the primary mechanism causing the difference?

CONSTRUCTED RESPONSE:

Using your wind chill model, explain why a marathon runner who finishes a race on a cold, windy day is at risk of hypothermia even though they were warm during the race. Reference convective heat loss, the thermal boundary layer, and how physical activity and wind speed interact in the system.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Wind chill is one of the most commonly experienced but least understood weather phenomena. Understanding the physics behind why wind makes cold feel colder reveals fundamental principles of thermal energy transfer and how the human body interacts with its environment.

The Thermal Boundary Layer

The human body constantly radiates heat, warming a thin layer of air directly against the skin. This thermal boundary layer acts as natural insulation, typically only a few millimeters thick. In still air, this layer remains intact and slows further heat loss. When wind blows, it strips this warm air away and replaces it with cold air, forcing the body to warm a new layer that is immediately stripped away again.

Convection: The Mechanism of Wind Chill

Convection is the transfer of thermal energy by the physical movement of a fluid. In wind chill, the fluid is air. Faster-moving air increases the rate of convective heat transfer because it replaces the warmed boundary layer more rapidly. This is the same principle that makes a convection oven cook faster than a conventional oven and why blowing on hot food cools it down.

Wind Chill Is Not Real Temperature

Wind chill is a calculated index that represents how fast exposed skin loses heat, not an actual air temperature. A wind chill of 0°F on a 20°F day with 25 mph wind means your skin loses heat at the same rate it would on a calm 0°F day. Objects that do not generate their own heat (like a car or a rock) will never cool below the actual air temperature regardless of wind.

Diminishing Returns at High Speed

The relationship between wind speed and wind chill is not linear. The first 10 mph of wind has a much greater cooling effect than going from 40 to 50 mph. This is because the thermal boundary layer is mostly disrupted at moderate wind speeds, so additional wind speed has less remaining insulation to remove.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify wind speed and air temperature as the two external components, and convective heat loss, skin temperature, and perceived cold as the three internal responses of the wind chill system.

E **Establish Relationships**

Students determine that wind speed positively increases convective heat loss, air temperature negatively affects convective heat loss (colder air means more loss), convective heat loss negatively affects skin temperature, and lower skin temperature positively increases perceived cold.

V **Visualize & Model**

Students build a model showing how wind disrupts the thermal boundary layer to increase heat loss, connecting wind speed and air temperature to the chain of convective heat loss, skin temperature drop, and perceived cold.

E **Evaluate Outcomes**

Students run calm vs. windy scenarios at the same temperature, then compare moderate wind at different temperatures to discover that wind amplifies cold but does not create it.

R **Revise & Extend**

Students add humidity level or clothing insulation to explore how additional factors modify the wind chill effect and protection strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"It is 35 degrees outside. On Monday with no wind, you are comfortable in a jacket. On Tuesday with 30 mph wind, you are shivering and your face hurts. Same temperature. What changed?"*

Watch For

Show weather reports from two days with same temperature but different wind. Let students compare.

Slide 2: Learning Objectives

3 min

Say: *"Today we are modeling why wind makes cold feel colder, and it is NOT because wind is cold. The answer is about how fast your body loses heat."*

Watch For

Have students read objectives. Pre-teach 'convection' and 'thermal boundary layer.'

Slide 3: Big Question

3 min

Say: *"A thermometer reads the same temperature whether wind is blowing or not. So if the air is the same temperature, why does your body disagree?"*

Watch For

Think-pair-share: When have you experienced wind chill? What did it feel like compared to a calm day?

Slide 4: LEVER Framework

2 min

Say: *"We are going to model the invisible layer of warm air your body creates around itself and what wind does to that layer."*

Watch For

Draw a person with a thin warm air layer. Show arrows of wind stripping it away.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external inputs we cannot control, and which respond inside the system?"*

Watch For

Guide sorting. Wind speed and air temperature are external. Convective heat loss, skin temperature, and perceived cold are internal responses.

Slide 6: Activity 2: Connections

8 min

Say: *"Wind speed does not change the air temperature. It changes how fast your body loses heat. That is the key connection."*

Watch For

Trace the chain: wind speed increases convective heat loss, which decreases skin temperature, which increases perceived cold.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test it. Same air temperature, different wind speeds. Watch what happens to skin temperature."*

Watch For

Students predict then run calm, moderate wind, and high wind scenarios. Compare perceived cold values.

Slide 8: Discoveries

5 min

Say: *"Wind does not make the air colder. It makes YOU colder by stripping away your body's natural warm air insulation."*

Watch For

Connect to real life: blowing on soup, car windshield defrost, why mittens work better than gloves.

Slide 9: STEM Challenge

12 min

Say: *"Design a face shield that keeps the thermal boundary layer intact against wind while still letting you breathe and see."*

Watch For

Provide materials (fabric swatches, foam, plastic sheets). Test with a fan and thermometer. Measure temperature behind the shield.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wind Speed, Air Temperature
- **INTERNAL:** Convective Heat Loss, Skin Temperature, Perceived Cold

Relationships

- **Wind Speed to Convective Heat Loss:** POSITIVE (+) — Higher wind speed strips the thermal boundary layer faster, replacing warm air with cold air more rapidly and increasing the rate of heat transfer away from the skin.
- **Air Temperature to Convective Heat Loss:** NEGATIVE (-) — Lower air temperature increases the temperature difference between skin and air, driving faster heat transfer. Warmer air reduces the gradient and slows heat loss.
- **Convective Heat Loss to Skin Temperature:** NEGATIVE (-) — Greater convective heat loss removes thermal energy from the skin faster than the body can replace it, causing skin temperature to drop.
- **Skin Temperature to Perceived Cold:** NEGATIVE (-) — Lower skin temperature triggers more cold receptors in the skin, increasing the sensation of cold that the brain perceives.
- **Wind Speed to Skin Temperature (indirect):** NEGATIVE (-) — This indirect path shows how wind speed ultimately drives down skin temperature through the convective heat loss mechanism, even though wind does not change air temperature.

Key Discoveries

- Wind does not change the actual air temperature but dramatically increases the rate of heat loss from the skin
- The thermal boundary layer acts as a thin blanket of warm air against the skin that wind strips away
- Wind chill is not a real temperature but a measure of how fast the body loses heat
- At higher wind speeds, the additional cooling effect diminishes because the boundary layer is already fully disrupted
- Wet skin loses heat even faster in wind because evaporation adds another cooling mechanism on top of convection

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Wind makes the air colder"	Wind does not change air temperature. A thermometer reads the same whether wind is blowing or not. Wind increases the rate at which warm objects lose heat to the air by disrupting the insulating boundary layer. The air is the same temperature; your body just loses heat to it faster.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Wind chill temperature is a real temperature"	Wind chill is an index that describes heat loss rate, not an actual temperature. It tells you how fast exposed skin would lose heat compared to a calm day. A wind chill of -10°F does not mean anything outside will cool to -10°F. Only objects that generate their own heat (humans, animals) are affected by wind chill.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More wind always means proportionally more cold"	The relationship between wind speed and cooling is nonlinear with diminishing returns. The first 10 mph of wind has a much greater effect than going from 40 to 50 mph because the boundary layer is already mostly disrupted at moderate speeds. The model shows this curve clearly.